Forensic Biometrics & Master Persona Audit

ALL IDENTITIES CLASSIFIED AS UNKNOWN SUBJECTS (PENDING ENROLLMENT)
Total Discovered: 1,652 Personas (SUBJ-01 to SUBJ-1652). Sighting tiers reflect statistical recurrence volume only.
GROUND TRUTH VERIFICATION: Unknown subject SUBJ-1652 verified biometrically distinct from SUBJ-02 (similarity = 0.1831).

1. Tier 1 Master Directory: High-Frequency Recurring Unknown Subjects (50+ Detections)

Subject ID	Forensic Status	Total Dets	Active Days	First Seen	Last Seen	Contacts	Risk
SUBJ-01	Unknown Person	29,598	29	2026-09-01 01:29	2026-09-30 23:32	**116**	**CRITICAL**
SUBJ-02	Unknown Person	166	12	2026-09-01 09:04	2026-09-30 08:51	**53**	**CRITICAL**
SUBJ-03	Unknown Person	104	2	2026-09-01 07:29	2026-09-11 07:46	**0**	**LOW**
SUBJ-04	Unknown Person	87	16	2026-09-01 07:29	2026-09-30 08:13	**0**	**LOW**
SUBJ-05	Unknown Person	86	2	2026-09-16 16:19	2026-09-23 11:21	**0**	**LOW**
SUBJ-06	Unknown Person	85	12	2026-09-02 08:36	2026-09-30 09:14	**0**	**LOW**
SUBJ-07	Unknown Person	81	3	2026-09-01 11:14	2026-09-21 11:22	**0**	**LOW**
SUBJ-08	Unknown Person	79	10	2026-09-01 11:14	2026-09-29 11:14	**0**	**LOW**
SUBJ-09	Unknown Person	79	1	2026-09-25 14:03	2026-09-25 14:15	**0**	**LOW**
SUBJ-10	Unknown Person	75	2	2026-09-13 19:40	2026-09-21 19:37	**0**	**LOW**
SUBJ-11	Unknown Person	69	9	2026-09-01 14:14	2026-09-30 09:13	**0**	**LOW**
SUBJ-12	Unknown Person	68	2	2026-09-07 17:51	2026-09-09 17:50	**0**	**LOW**

2. Sighting Frequency Distribution (Tiers 1, 2, 3)

- **Tier 1 (High-Frequency Unknowns, 50+ Detections): 12 individuals** (30,577 detections)
- **Tier 2 (Medium-Frequency Unknowns, 10 - 49 Detections): 85 individuals** (1,701 detections)
- **Tier 3 (Transient Unknowns, 2 - 9 Detections): 1,555 individuals** (3,948 detections)
- **Total Discovered Registry: 1,652 individuals** (36,226 total face detections)

3. Critical Equipment Protection Zone Breach Ledger (18 Unknown Individuals)

Across the 30-day archive, exactly **18 unknown individuals** established verified direct physical contact (IoU > 0.08) with plant equipment assets:

Subject ID	Contacts	Active Days	Total Dets	Primary Equipment Asset	Hazard
SUBJ-01	**116**	29	29598	Two Multiport Backwash Valves	**CRITICAL**
SUBJ-02	**53**	12	166	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-102	**5**	1	8	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-167	**5**	1	6	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-41	**4**	10	20	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-344	**3**	1	3	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-395	**3**	1	3	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-129	**2**	1	7	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-489	**2**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-22	**1**	1	35	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-342	**1**	1	3	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-582	**1**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-759	**1**	1	2	Ultimate Plant Power Switch	**SEVERE**
SUBJ-842	**1**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-890	**1**	1	2	Two Multiport Backwash Valves	**CRITICAL**
SUBJ-993	**1**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-1079	**1**	1	2	Ultimate Plant Power Switch	**SEVERE**
SUBJ-1652	**1**	1	28	Two Multiport Backwash Valves	**CRITICAL**

4. Comparative Forensic Analysis: Unknown Subject SUBJ-1652 vs SUBJ-02

High-resolution facial feature analysis confirmed that the unknown individual observed operating the multiport valve on Day 1 at 09:53:38 is **distinct from SUBJ-02**:

Biometric Attribute	Unknown Subject SUBJ-02 (09:04:32)	Unknown Subject SUBJ-1652 (09:53:38)	Forensic Verdict
ArcFace 512-D Cosine Sim	Baseline (1.0000)	**0.1831**	**NON-MATCH (Threshold >= 0.4500)**
Normalized Correlation	Baseline (1.0000)	**0.0434**	Zero correlation
Attire & Visuals	Dark Navy / Black Shirt	Teal / Green Polo Shirt	Distinct work attire
Facial Morphology	Straight hair, slim jawline	Thick curly hair, mustache & goatee	Distinct facial structure
Incident Timestamp	09:04:32 (EVT-D01-003)	09:53:38 (Day01_0953)	Independent morning access